

EMBODIED ATTENTION BUDGETING

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ABSTRACT

Active perception usually asks which observation would be most informative. For embodied robots, that question is incomplete: sensing consumes time, physical configuration, compute, power, and safety margin. We propose *embodied attention budgeting*: a controller treats sensing as a limited physical action and spends it only when the expected control-risk reduction justifies the budget. A small hazard-navigation simulation shows why the distinction matters. A no-sensing baseline reaches the goal often but collides frequently. A greedy information policy over-senses and misses the finite control horizon. A risk-coupled budgeted policy spends only two sensing actions, reduces collisions, and keeps enough horizon to finish the task. The result is not a broad benchmark claim; it is mechanistic evidence that attention should be accounted for as an embodied resource, not treated as free uncertainty reduction.

1 INTRODUCTION

Robots do not observe from nowhere. A camera view can require motion, a tactile measurement can require contact, and a high-resolution perception pass can cost latency while the robot is still moving. Yet much active perception work frames the problem as choosing the most informative observation. That framing hides a physical assumption: sensing is cheap enough, safe enough, and separable enough to optimize before or outside control.

This paper asks a narrower question. What if attention itself is a scarce embodied resource? We use *attention* broadly: visual fixation, tactile probing, active view selection, high-resolution inference, or any sensing action that consumes time and physical opportunity. The contribution is not a larger model, a new benchmark, or reinforcement learning. It is a change in accounting. The robot should spend attention only when the sensing action preserves more control value than it consumes.

Our claims are modest. First, information gain alone can overvalue sensing when there is a finite control horizon. Second, a risk-coupled budget can select less information and still produce better task behavior. Third, the difference is observable in a small runnable task with a fixed hazard, a finite horizon, and a limited attention budget.

2 RELATED WORK

The closest areas are active perception, active sensing, sensor scheduling, information gathering, tactile exploration, and perception-aware manipulation. The 2,013-row sweep in `docs/related_work_matrix.csv` covers robotics and robotics-adjacent arXiv papers, then narrows to active-perception and sensor-management clusters. The strongest hostile priors include active sensing surveys, multi-robot information acquisition, attention-based planning with active perception, submodular active perception, and active neural sensing.

These works make it non-novel to choose views, maximize information, schedule sensors, or couple perception to action. The remaining gap is the objective. Many methods ask whether an observation is informative. Fewer ask whether the robot can afford the embodied act of attending under immediate control risk.

3 BUDGETED ATTENTION

Let s_t be the robot state, b_t its belief uncertainty, u_t a control action, and o_t a sensing action. A standard active-perception objective can choose o_t by maximizing expected information gain. We instead introduce an attention budget B_t and a risk-coupled spending rule:

$$o_t \text{ is allowed only if } \Delta R(s_t, b_t, o_t) > C_{\text{attn}}(o_t, B_t, H_t),$$

where ΔR is expected control-risk reduction, C_{attn} is the embodied cost of sensing, and H_t is the remaining control horizon. The important feature is not the exact formula; it is that attention competes with actuation for finite physical opportunity.

4 TOY EVIDENCE

We implement a one-dimensional hazard task in `scripts/toy_attention_budget.py`. The robot must reach a target at position 10 within 12 steps while crossing a hazard near position 5. Moving under uncertainty can collide. Sensing reduces uncertainty, but it consumes one step and one unit of attention.

The policies are deliberately simple. The baseline never senses. The greedy policy senses until uncertainty is very low. The budgeted policy senses only while uncertainty is high and the robot is not at the risky hazard boundary. All policies are evaluated for 2,000 deterministic seeds.

Policy	Success	Collisions	Avg. return	Attention	Info
Baseline	836	1164	-1.734	0.000	0.000
Greedy	0	114	-1.767	5.000	3.564
Budgeted	1572	428	1.049	2.000	-0.044

Table 1: Toy attention-budget task. Greedy sensing gathers more information but spends too much horizon to finish. Budgeted attention spends less and succeeds more often.

The exact values are less important than the failure mode. Information gain is not the right objective when sensing consumes the same finite horizon needed for safe control.

5 LIMITATIONS

This is a toy experiment, not a hardware result. The budget rule is hand-coded, the hazard is simple, and the attention cost is stylized. The claim is also close to prior work in active perception with costs. The novelty boundary is therefore narrow: attention is treated as a physically consumed control resource, not merely as an observation-selection cost.

6 CONCLUSION

Embodied attention should be budgeted. In robots, sensing can improve belief while still damaging control opportunity. A small simulation shows that a risk-coupled budget can outperform both no sensing and greedy sensing. The next step is to validate the same accounting on real perception-action loops where view changes, tactile probes, and compute latency have measurable physical cost.

REFERENCES

Active sensing for robotics surveys; decentralized active information acquisition; attention-based planning with active perception; submodular active perception; multi-modal active perception for information gathering; active perception and representation for robotic manipulation; robot active neural sensing and planning; tactile and visuo-tactile active perception; and perception-aware robot navigation and manipulation papers identified in `docs/hostile_prior_work.md`.